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Integrated Project: End-to-End Solution

Python Pathway | Session 8

Session Recording & Professional Expectations

Session Recording

- All Estio training sessions are recorded to:
 - Support learner progress
 - Enable review of content
 - Maintain quality and compliance standards
- By attending this session, you acknowledge that recording is taking place.

Camera & Participation Expectations

- To support engagement and attendance monitoring:
 - Cameras should remain on during the session
 - Please participate actively in discussions and activities
 - If you are unable to use your camera or microphone for any reason, please inform your tutor directly via chat or message.

This session forms part of your professional apprenticeship experience. Please engage respectfully and professionally throughout.

Safeguarding, British Values & Wider Learning

Safeguarding protects individuals from harm, abuse, neglect, radicalisation and exploitation.

If you have a concern about yourself or someone else, email:

safeguarding@bpp.com

Or speak to your tutor privately.

We treat all concerns seriously and confidentially.

Safeguarding



The British Values are:

1. **Democracy**
2. **Rule of Law**
3. **Individual Liberty**
4. **Mutual Respect**
5. **Tolerance of Different Faiths and Beliefs**

These values underpin everything we do in professional and digital environments.

British Values



Prevent is about protecting people from being drawn into extremism or radicalisation.

In our programmes, Prevent links to:

- **Online safety**
- **Digital responsibility**
- **Critical thinking**
- **Professional ethics**

Prevent



As we go through today's session, consider:

How today's topic connects to British Values

Whether there are any Prevent or digital safety themes

How this learning supports your professional and career development

Your tutor will guide discussion and help you make these connections.

Competencies and Coverage

Professional Discussion underpinned by Portfolio

Knowledge

K1, K3, K4, K5, **K7**, K8,
K10, K12

Skills

S2, S3, **S5**, S8, S9, S13,
S14, S15, S17

Behaviours

B1, B4, B5, B6, B7, B8,
B9

Work-based Project with Questioning

Knowledge

K2, **K6**, **K9**, K11

Skills

S1, **S4**, **S6**, **S7**, S10,
S11, S12, **S16**

Behaviours

B2, B3

Lesson Aim and Objectives

Lesson Aim: Integrated Project: End-to-End Solution

Lesson Objectives

Create

- Create a complete working program applying core programming principles

Evaluate

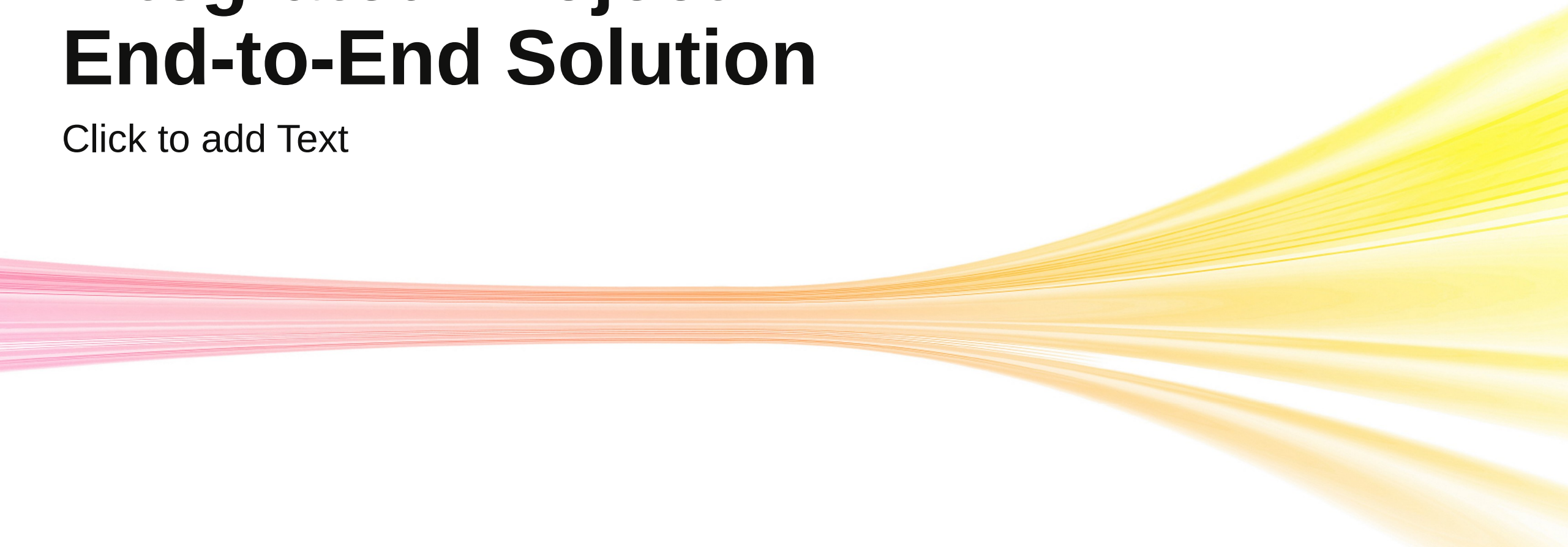
- Evaluate design decisions and justify chosen approaches

Apply

- Apply testing, documentation and structured problem-solving techniques

Integrated Project: End-to-End Solution

Click to add Text

A decorative graphic consisting of multiple thin, overlapping wavy lines that create a sense of motion and flow. The lines originate from the left side of the slide, where they are primarily pink and light orange, and extend towards the right, where they transition into a vibrant yellow. The lines vary in thickness and opacity, creating a layered, ethereal effect.

The Workplace Data Processor

From Structured Design to Functional Implementation



Employee Timesheet & Payroll Processor

Scenario Description:

You must build a command-line tool for managing employee weekly hours.

The system processes records, flags anomalies, calculates pay, and generates reports.

Data Constraints

- Hourly base rate is £25.00
- Overtime is paid at 1.5x for hours over 40
- Maximum allowable hours per week is 60





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Phase 1

Project Briefing & Planning

System Requirements & Logic Design

- Define explicit requirements from the payroll scenario
- Map input data, processing formulas, and output report layouts
- Structure program logic using visual flowcharts or pseudocode
- Align development milestones with standard SDLC planning stages
- Deliver verifiable blueprints mapping data transformations comprehensively

Deliverables & Action Items

Planning Phase Tasks & Outcomes

- **Task 1:** Extract and define explicit requirements from the scenario
- **Task 2:** Document exact data inputs, processing loops, and output schemas
- **Task 3:** Create verified structural designs using pseudocode/flowchart blueprints

Expected Outcome:

A comprehensive, structured design blueprint outlining clear program logic



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Phase 2

Solution Development

Technical Specifications & Mechanics

- Accept manual data entries via keyboard inputs
- Validate entered values against established domain boundaries
- Process records using iteration loops and conditional statements
- Store in-memory records using optimised primitive data structures
- Implement searching or sorting algorithms across active datasets
- Render clear, human-readable summaries to the screen



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Phase 3

Structured Implementation
Expectations



Clean Coding & Modularity Standards



- Break monolithic logic blocks into scoped, single-purpose functions
- Enforce strict, consistent variable and function naming conventions
- Maintain clean layout formatting and standard indentation patterns
- Approach debugging tasks using systematic, structured problem-solving steps
- Deliver highly readable, modular, and maintainable source code

Development Milestones

Development Milestones & Outcome

- **Integration Goal:** Merge distinct algorithmic concepts into a single cohesive solution
- **Input Control:** Ensure data entry blocks invalid values automatically
- **Algorithm Test:** Verify sorting/searching features execute with minimal complexity

Expected Outcome:

A fully functioning software program meeting all integration specs



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Phase 4

Testing & Validation



Quality Assurance & Defect Tracking

- Design test scenarios covering diverse operating data conditions
- Include standard nominal values, boundary edges, and invalid entries
- Execute test matrices and capture all runtime behaviours systematically
- Track down, isolate, and debug underlying logic errors
- Document comprehensive logs proving execution refinement and system stability



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Phase 5

Collaboration & Review

Peer Feedback & Optimisation Loops

- Participate in small group peer code review sessions
- Evaluate external source files for readability and logical clarity
- Offer objective, actionable, and constructive technical feedback points
- Refactor local software solutions incorporating received peer commentary
- Showcase collaborative iterations considering alternative computational approaches

Peer Review Framework

Group Collaboration & Review Workflow:

- **Peer Review:** Form small groups to review peer code repositories
- **Evaluation Metrics:** Inspect source files for readability, logical clarity, and structure
- **Objective Critique:** Deliver constructive, actionable feedback points on code layout
- **Refactoring:** Incorporate the recommended design enhancements back into your local project

Core Objective: To build improved solutions that show active consideration of alternative coding approaches

Code Review Checklists

Code Readability & Logic Benchmarks:

- **Structural Clarity:** Check if functions have a single, modular responsibility
- **Naming Standards:** Verify variable and function names match established code styling
- **Complexity Check:** Identify redundant loops, deeply nested conditions, or unreachable statements
- **Error Resilience:** Confirm validation blocks intercept type conflicts or boundary errors
- **Documentation Depth:** Ensure complex algorithms include concise descriptive docstrings and inline comments



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Phase 6

Presentation & Justification

Solutions Engineering Defence

- Pitch completed software solutions to project stakeholders cleanly
- Explain chosen architectural styles and layout structural designs
- Trace real runtime logic paths to demonstrate functional success
- Defend engineering trade-offs regarding memory use and speed
- Identify future optimisation opportunities for subsequent releases

Presenting the Final Solution

Technical Presentation Expectations:

- **Solution Pitching:** Present completed, running software applications to target audiences clearly
- **Engineering Defence:** Justify design trade-offs regarding speed, memory, and complexity
- **Design Walkthrough:** Explain architectural design decisions and the project's overall structure
- **Continuous Improvement:** Outline future software optimisations based on evaluation metrics
- **Logic Verification:** Describe exactly how core functionality was written and successfully implemented
- **Core Objective:** Explicitly articulate design decisions and prove a deep understanding of software quality

Build, Test & Present Your Solution



ACTIVITY



Objective: *Design, Build, Test & Present* a working "Workplace Data Processor"

Tasks / Requirements:

- Capture and validate employee hours input
- Process data using loops, conditionals and an appropriate data structure
- Implement a sort or search feature on the stored records
- Produce a clear, formatted output report

Key Point: Be ready to justify your design choices and identify any strengths / limitations

Extension Activity: Add functionality beyond the minimum requirements, e.g. category filtering



ACTIVITY



Stretch and Challenge

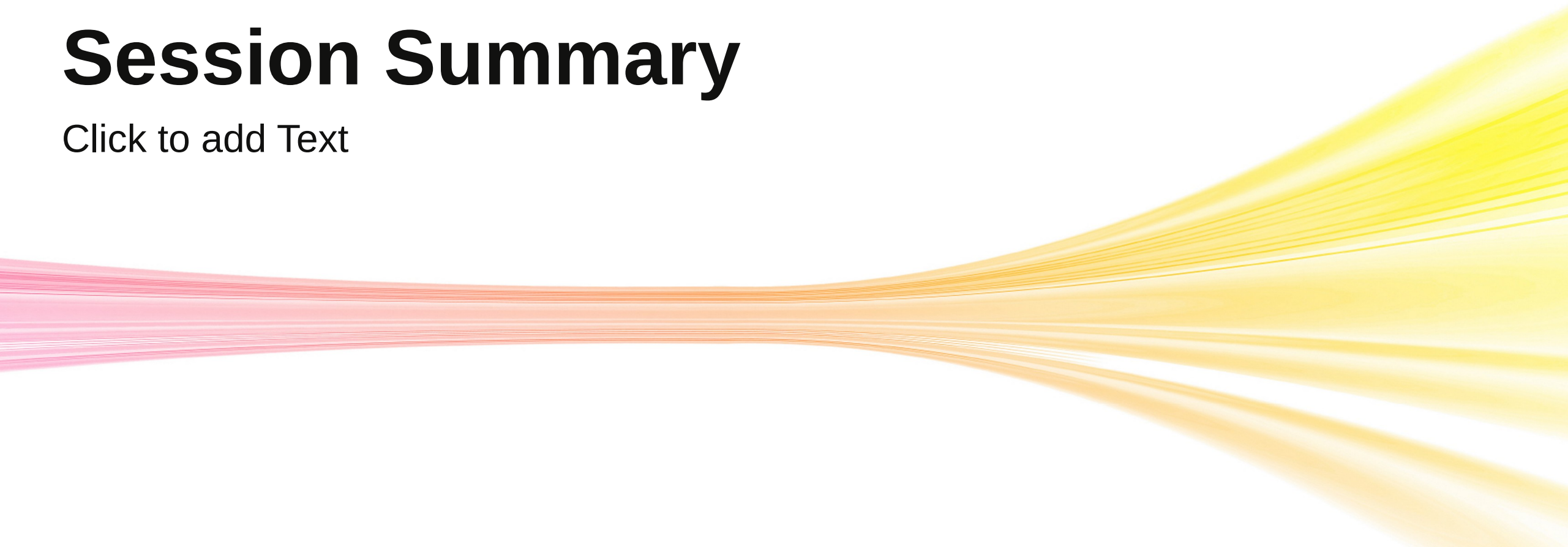
The Mission:

Upgrade your Workplace Activity Processor into production-grade code

- **Extended Features:** Add bonus business metrics like tax calculations or automated bonus brackets
- **Persistent Storage:** Implement file saving and loading via JSON or CSV file formats
- **Advanced Searching:** Code binary or linear searches to look up employees by explicit names
- **Optimised Sorting:** Refactor baseline lookups using custom algorithmic keys or secondary sort fields
- **UI Refinements:** Build a persistent, clean loop-driven terminal menu system for easy navigation

Session Summary

Click to add Text



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